

Honeywell

Fieldbus Fundamentals

FB01-610

R610

08/00

TotalPlant

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Release 610 August 31, 2000**

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About This Document

This document describes Fieldbus system architecture and wiring and installation.

Release Information

Document Name	Document ID	Release Number	Publication Date
Fieldbus Fundamentals - FB01	FB01-610	610	08/00

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE - EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.

Symbol	Definition
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.
	CAUTION: Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices. CAUTION symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING: Indicates a potentially hazardous situation which, if not avoided, could result in serious injury or death. WARNING symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING, Risk of electrical shock: Potential shock hazard where HAZARDOUS LIVE voltages greater than 30 Vrms, 42.4 Vpeak, or 60 VDC may be accessible.
	ESD HAZARD: Danger of an electro-static discharge to which equipment may be sensitive. Observe precautions for handling electrostatic sensitive devices
	Protective Earth (PE) terminal: Provided for connection of the protective earth (green or green/yellow) supply system conductor.
	Functional earth terminal: Used for non-safety purposes such as noise immunity improvement. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national local electrical code requirements.

Symbol	Definition
	Earth Ground. Functional earth connection. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.
	Chassis Ground: Identifies a connection to the chassis or frame of the equipment shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.

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1. Introduction to Fieldbus

1.1 What is Fieldbus?

Fieldbus represents a concept designed to meet today's plant needs for tighter integration of control and field devices. Through the use of a standard digital communications protocol, intelligent devices can share values, parameters, display and trending information, and alarms and events with the control room and other field devices. Because Fieldbus is a multi-drop communications network, you can significantly reduce wiring costs.

Introductory Fieldbus concepts

The following are introductory Fieldbus concepts presented in this section:

- Fieldbus definition
- Fieldbus is an enabling technology
- Fieldbus represents more than a communication standard
- Not all buses are Fieldbus

Fieldbus definition

Fieldbus is defined as an “all digital, serial, two-way communications system that interconnects measurement and control equipment such as sensors, actuators and controllers. At the base level in the hierarchy of plant networks, it serves as a Local Area Network (LAN) of instruments used in process control and manufacturing automation applications and has a built-in capability to distribute the control application across the network.” (Source: Fieldbus Foundation at <http://www.fieldbus.org>)

Fieldbus is an enabling technology

Fieldbus is often described as an enabling technology because it enables

- an intelligent digital replacement for the traditional 4-20ma approach, and
- unifies field devices with business and/or control automation systems.

Fieldbus represents more than a communication standard

The most common misconception about Fieldbus is that it is just a communication standard. Fieldbus represents much more than just a standard. Because Fieldbus supports a configurable user layer in its architecture, it has many of the characteristics of a process control system. Fieldbus defines a system of function blocks that can be resident in the field devices and interconnected as a distributed process control system. A function block is very similar to a controller's control slot. A function block executes an algorithm operating on configured input(s) and parameters and produces an output.

Not all buses are Fieldbus

Many device buses used in plants are being called Fieldbus, creating confusion among users. Although there are several types of buses - -such as sensor buses, device buses, and local area networks - - the network designs have different purposes. The term Fieldbus properly applies only to buses with technical correspondence to the IEC/ISA SP50 Fieldbus standard, such as Foundation Fieldbus.

The Fieldbus Foundation, a not-for-profit consortium of over a hundred instrumentation and control companies, defined the specification FOUNDATION™ Fieldbus. The FOUNDATION™ Fieldbus specification is not controlled through any individual company, nation, or standards body.

Look for this symbol!

How do you determine whether a field device that you wish to purchase complies with the Fieldbus standard? One way is to look for the Foundation™ Fieldbus 'seal of approval.'



The Fieldbus Foundation has a registration program whereby control instrument suppliers send their devices in for testing. Devices that pass the test can attach the Fieldbus Foundation seal of approval. The Fieldbus Foundation's conformance testing ensures that devices are compliant and interoperable. The registration tests do not include host systems, configuration tools, and other Fieldbus related devices.

Rationale for Fieldbus usage in plant networks

In order to understand the rationale for a Fieldbus system, recall that process control evolved from monitoring at a local level, to centralized control within the plant, then to the current era distributed process control. Fieldbus represents the next step in the evolution of process control and monitoring architecture. The Fieldbus digital communications link can interconnect the sensors and actuators to the control room, control can be located local to the devices, and the digital devices can communicate directly to each other and fully exploit each other's capabilities. The concept is called "fieldbus" control and monitoring.

1.2 How does Fieldbus Benefit Plant Operations?

Key Fieldbus benefits

Fieldbus benefits plant operations in the following areas

- Installation
- Operation
- Maintenance
- Interoperability

All of these cost saving benefits allow your plant to become more competitive.

Installation benefits

Installation benefits include lower wiring costs, remote configuration, and multiple suppliers. There is less wiring because of the multi-drop capability. Fieldbus access to parameters of the device and its function blocks permit a device to be configured remotely, over the wire. This means that configuration software can perform legitimacy checks, prepare a complex database, and automatically document the configuration.

Operation benefits

Operational benefits include higher accuracy, more precision, better control, safer control, more information, reliable and secure operation, and less downtime. Digital transmission avoids the losses in accuracy caused by analog conversion. IEEE floating point representation provides seven digits of precision, much more than with analog transmission. The ability to detect a bad signal yields better and safer control. Transmission security checks ensure detection of transmission errors and recovery techniques ensure reception of data or at least the knowledge that data was lost. The increased reliability of digital circuits results in less downtime than their corresponding analog circuits.

Maintenance benefits

Maintenance benefits include increased reliability, remote diagnostics, and competitive replacements. Remote access to parameters will permit devices that diagnose themselves to be able to present the diagnostic information over the wire and even to provide alarm and event notification where needed. This will save many trips to locations that are high, hot, wet, dangerous, hard-to-get-to, or otherwise inaccessible.

Interoperability

Since the standard ensures interoperability, when a Fieldbus device needs to be replaced, you select the best device for an application.

1.3 Fieldbus Differentiators

All vendors are not the same!

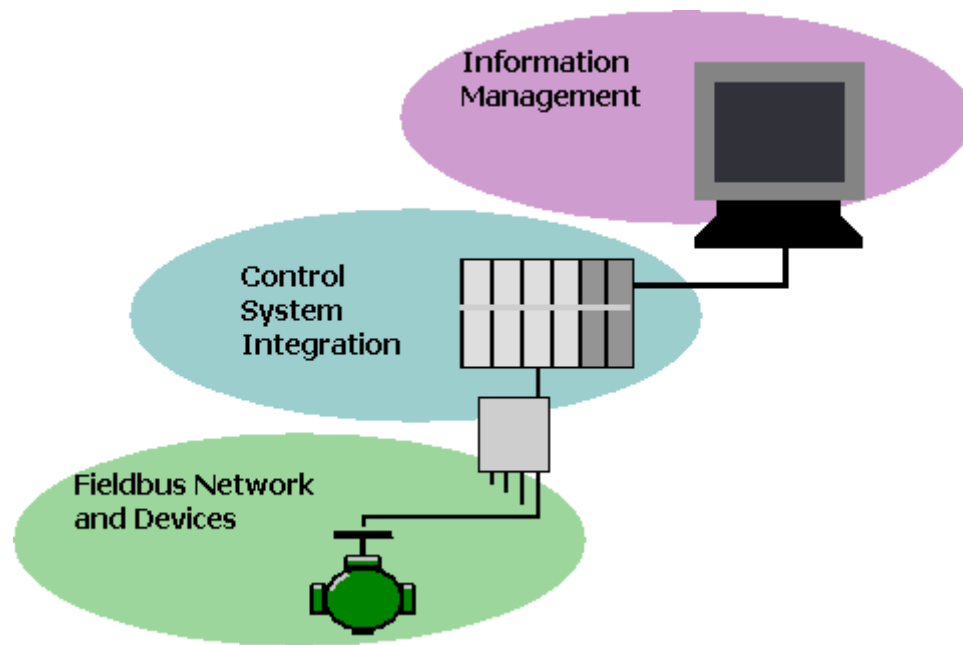
If Fieldbus defines so much, what is left to differentiate the vendors? Here are some of the things not addressed by the standard: superior measurement and actuator technology; functions beyond the standard; application expertise; price; the quality of sales support, service, and the product; and maintainability of the product.

Initial customer response to Fieldbus

Many users recognize that Fieldbus is a dramatic new technology. Hence, initial Fieldbus applications are often small and non-critical. As users become familiar with the technology and recognition of its benefits increase, applications of Fieldbus products will grow. Users will accept that Foundation Fieldbus is not just a standardized multi-drop digital communications standard, it is an enabling technology and a new architecture for control and monitoring.

Honeywell's approach: OpenField

Honeywell is the most experienced system supplier when integrating digital field devices. Honeywell's approach, OpenField, extends a control system to include Fieldbus devices. An integrated system provides information management to the business domain and security for the process domain. Instrument records, Fieldbus configurations, predictive diagnostics can become part of your information management strategy. Control systems can integrate process control, protect databases, and support consistent operator displays. Fieldbus networks become interoperable, with local control and associated installation savings.



2. Fieldbus System Architecture

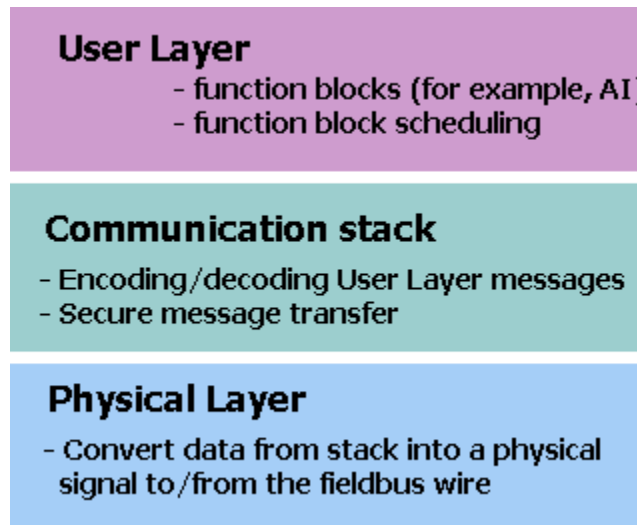
2.1 Fieldbus Communication Model Overview

Key communication concepts

The most important Fieldbus communication concept is that the devices must use a common language or no meaningful communications will take place. The Fieldbus protocol and services are simply a mechanism to assure reliable digital communications between two devices or between a field device and a host control system. The Fieldbus communication model acts similar to sending a message in a letter through a postal system.

Communication model simplified

The Fieldbus communication model is frequently the topic of discussion in Fieldbus presentations and articles. The Fieldbus communication model is the mechanism around which the Fieldbus standards have a relationship to the Open System Interconnect (OSI) communication model. (The OSI model is discussed later.) The following figure shows three functional areas that make up the Fieldbus communication model – the User Layer, the Communications Stack, and the Physical Layer.

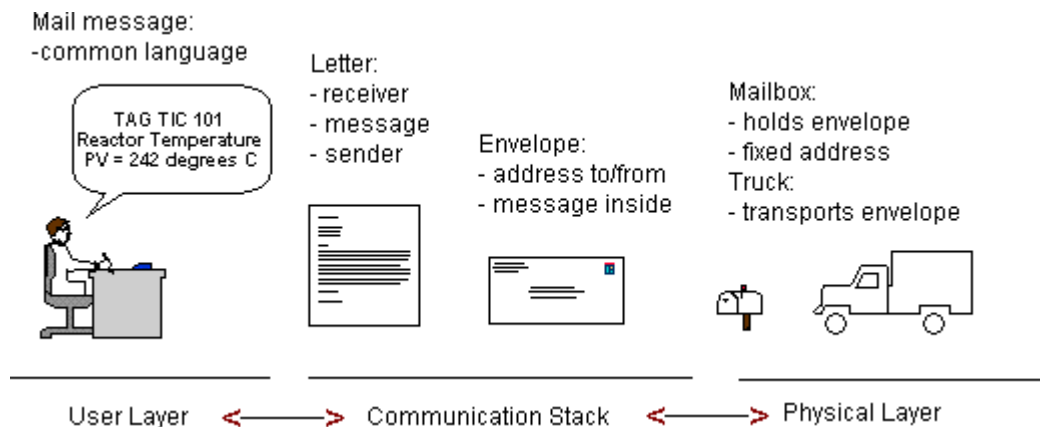


The User Layer in Fieldbus represents the common communications language, the element of Fieldbus that enables meaningful communications to take place. The User Layer provides standard function blocks (for example, an Analog Input (AI) function block), the scheduling of the function blocks, and host control system operability. The

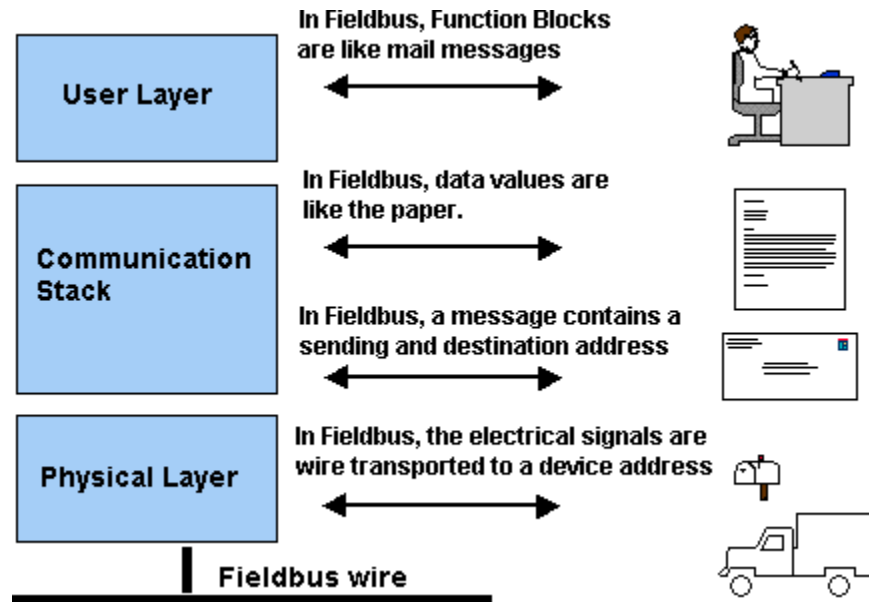
Communication Stack corresponds to OSI layers that encode and decode User Layer messages and secure their transmissions. The Physical Layer converts data from the Communication Stack into the 31.25 kbit/sec physical signal to (and from) the fieldbus wire.

Fieldbus acts like a postal system

The Fieldbus communications model acts as if it were a postal system. Simply stated the Fieldbus message must be written in a common language, contained in letter and envelope framework, and transported across a highway.



Below the User Layer, the Communications Stack takes care of putting a message on a piece of paper. The paper then becomes important, not the message itself. The paper is then placed in an envelope. The envelope then becomes important, not the paper or the message. The message is then placed in the mailbox, a flag set to indicate delivery, and eventually a mail truck comes by to pick it up. At this point it doesn't matter what route the mail takes, or what post offices it is routed through. The postmaster coordinates pickup and delivery. The message is transported independent of the message on the paper. Finally the envelope arrives at its destination. It is carried to the destination to which the envelope was addressed. When the person reads the message, meaningful communications will take place, but only if it is written in a common language. The message may contain information for the user at the destination, or it may request information from the user at the destination (or both).



2.2 Fieldbus Device Architecture

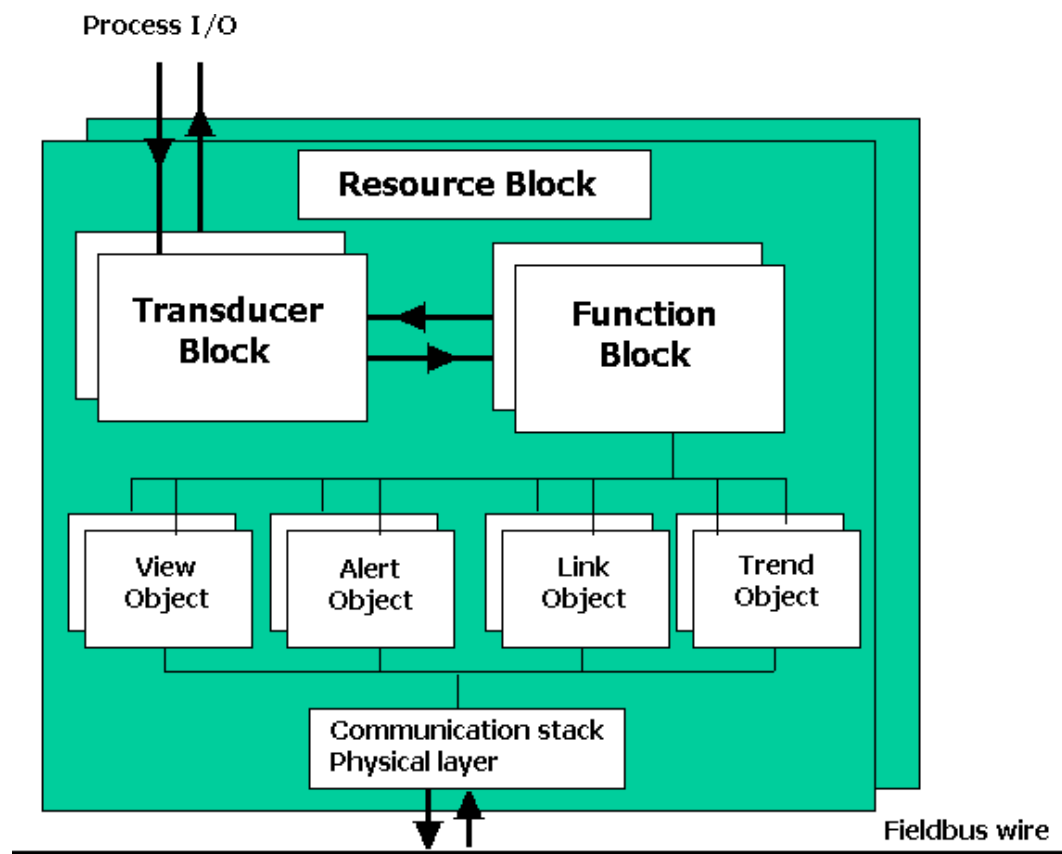
Key device architecture concepts

The key concepts about Fieldbus device structure are that a Fieldbus device consists of the following:

- one or more Function Blocks,
- a Resource Block,
- a Transducer Block, and
- Link, Trend, Alert and View objects.

Device structure

A Fieldbus device structure can be easily explained in terms of its blocks - - function, transducer, and resource - - and objects – view, alert, link, and trend. There's much more to a field device, of course, but this discussion provides a functional overview. A device structure in terms of blocks and objects is shown in the following figure.



The Fieldbus specification provides a user-layer based on blocks, which represent the different application functions. These blocks provide the system with standard information by which a device is configured, scheduled, operated and maintained. Control strategies are simplified through the use of standard function blocks for configuration.

What is a function block?

A function block is defined as an executable software object that performs a specific task, such as measurement or control, with inputs and outputs that connect to other entities in a standard way. Function blocks define the control behavior and link the input/output parameters to an algorithm. There can be several function blocks in a user application.

The standard function blocks are listed in the following table.

Function Block Name	Symbol
Analog Input	AI
Analog Output	AO
Bias	B
Control Selector	CS
Discrete Input	DI
Discrete Output	DO
Manual Loader	ML
Proportional/Derivative	PD

What is a resource block?

A resource block defines the fieldbus device. A resource block is defined as a named organization of parameters, which provides an interface to common resource capabilities or information. A resource block contains common information such as device type, device revision, and manufacturer's name. Common resource block capabilities include Fault State and Shed Options.

What is a transducer block?

A transducer block is defined as a named entity that specifies the required parameters associated with the production of an input value from a sensor or an output value to an actuator. In other words, transducer blocks provide the interface between a manufacturer's specific I/O and standard function blocks. A transducer block contains

information about sensor type or actuator type data and calibration data. Usually, each input or output function block has an associated input or output transducer block. The standard transducer blocks are listed in the following table.

Transducer Block Name
Standard Pressure with Calibration Basic Device Access
Standard Temperature with Calibration Basic Device Access
Standard Radar level with Calibration Basic Device Access
Standard Flow with Calibration Basic Device Access
Standard Basic Positioner Valve Basic Device Access
Standard Advanced Positioner Valve Basic Device Access
Standard Discrete Positioner Valve Basic Device Access

What are link, trend, alert and view objects?

Additional objects within a Fieldbus device include the following:

- Link objects represent the connections between function block inputs and outputs internally to the device and externally across the Fieldbus network.
- Trend objects support trending of function block parameters.
- Alert objects support alarm and event reporting.
- View objects are predefined groups of parameters viewable from an operator station.

2.3 Fieldbus Function Blocks

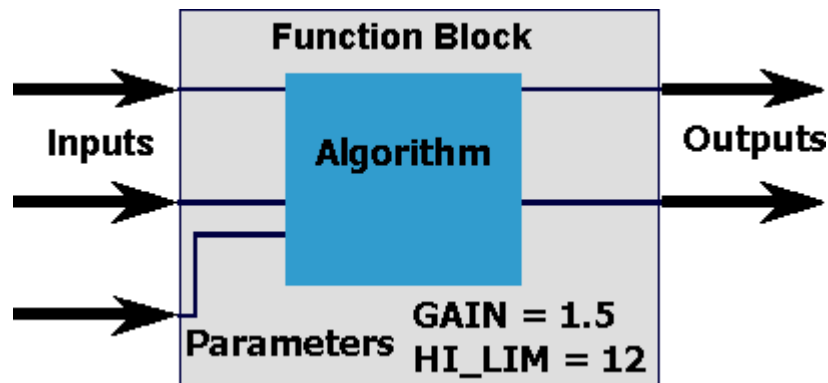
Key function block concepts

The following are key concepts about Fieldbus Function Blocks

- Function blocks execute a control behavior based upon their inputs, outputs, algorithm, and parameters.
- Function blocks fall into one of four classes - input, output, control, or calculation.
- Function blocks are executed upon a scheduled basis.

How a function block works

If you are familiar with control loops, you have an idea about how a function block works. Just like a control loop, a function block works using inputs, outputs, algorithms, and parameters. A function block overview appears in the following figure.



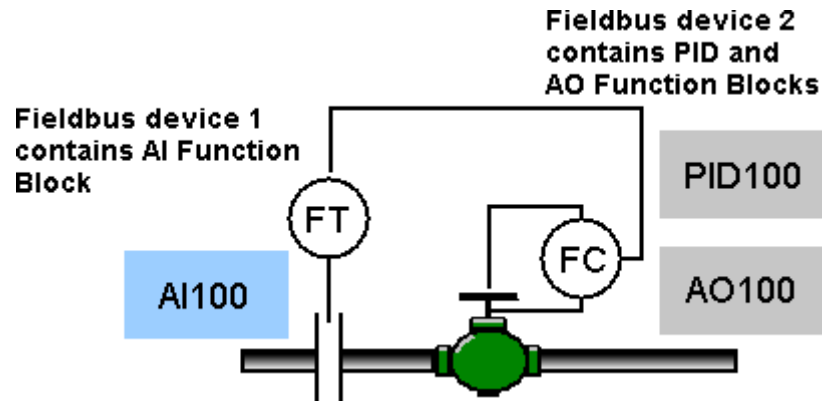
Four classes of function blocks

Function blocks are categorized into four classes:

- Input class, examples include analog, discrete, and pulse input.
- Output class, examples include analog and discrete output.
- Control class, examples include PID, lead-lag, and ratio control.
- Calculation class, examples include input selector, integrator, and signal characterization.

Example Fieldbus control loop operation

Consider the Fieldbus implementation of an example control loop. Assuming that you have an Analog Input (AI) function block in Device 1 and PID and Analog Output (AO) function blocks in Device 2, a control loop could be arranged on the Fieldbus as shown in the following figure. The AI function block is contained within the flow transmitter device 1. The PID and AO function blocks are contained within the actuator device 2. At a scheduled time interval, Device 1 broadcasts its AI value over the Fieldbus network. Device 2 receives the AI value for its PID algorithm.

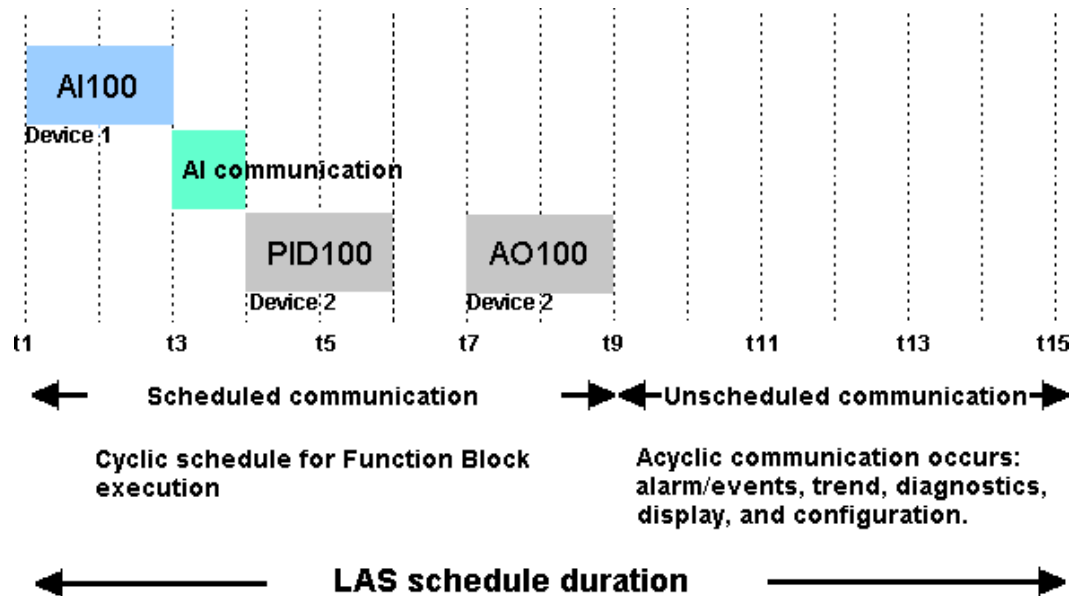


Scheduling synchronizes function blocks

For Fieldbus to provide process control at the field device level means that some scheduling function must synchronize the function blocks at precise periodic execution intervals. Additionally, field devices must be able to report high priority alarm events, display information, diagnostic, and trend information. The cyclic communication needs for process control and acyclic communication needs for events are met through the use of a deterministic scheduler called the Link Active Scheduler (LAS). Honeywell provides a schedule building tool that generates the schedules for the cyclic function block execution.

Conceptual view of a link schedule

Next, review the following figure that presents a conceptual view of a link schedule (The Link Active Scheduler is discussed in more detail later in this document.) The example schedule uses the earlier control loop figure to help illustrate some basic scheduling concepts.



In the conceptual figure of a user defined LAS schedule, all Fieldbus devices are aware of the link schedule start time (shown as $t1$ in our example). Observe that Device 1's AI100 begins its AI function block execution with a user configured zero time offset from $t1$. At time interval offset $t3$, the AI value is broadcast or "published" on the network. At time interval offset $t4$, the AI value is received or "subscribed to" by Device 2 for its PID function block. In other words, function block execution occurs on a scheduled basis as a user-configured offset of time from the scheduled LAS start time.

Scheduled time is for cyclic communication

Scheduled time is intended for your function blocks to execute. In other words, function block execution occurs on a user configured scheduled basis as an offset time interval from LAS absolute start time.

Unscheduled time is for acyclic communication

Next observe that when unscheduled time is available in the LAS schedule, acyclic communication occurs for alarms, trends, diagnostics, displays, and configuration. The LAS schedule then repeats itself at $t1$.

Having described how Fieldbus communicates from a functional overview, you can now consider additional details about Fieldbus communication.

2.4 Relationship to OSI Model

Key OSI concepts as they relate to Fieldbus

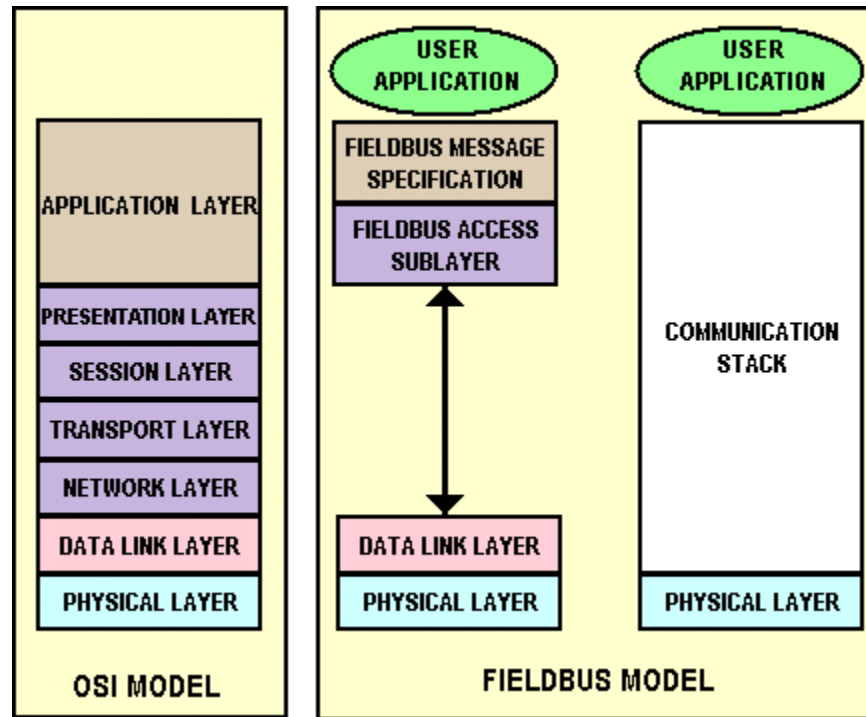
An earlier discussion indicated that for most Fieldbus implementers, the User Layer, which contains your function blocks, is the main layer to be concerned about. The remaining layers take care of transferring the necessary values and schedules for your Fieldbus strategy. The discussion that follows provides background on terms that you may encounter in various Fieldbus literature. The key concepts to become familiar with include the following.

- A Fieldbus device that issues the LAS schedules is called the Link Master.
- The terms publish and subscribe represent broadcast and receive.
- The Fieldbus technology model relates to the OSI model in several, but not all, OSI layers (Application, Data Link, and Physical).

What is an OSI model?

The Fieldbus technology model is based upon the Open Systems Interconnect (OSI) model. The OSI model provides a framework within which program interfaces are established. You may have seen the OSI model presented in computer networking discussions as a seven-layer model.

It's not always necessary for a particular network communication technology to use all of the seven layers simply because the functions several of the layers provide are not needed. For the Fieldbus model, the Presentation, Session, Transport, and Network layers are not needed. What does apply to the Fieldbus model from the OSI model are the Physical Layer, the Data Link Layer, and the Application Layer.



Role of Physical Layer

The Physical Layer is responsible for the actual signal transmission over the network. It receives the physical signals from other network devices and converts the signal to a message for the upper layer. The Physical Layer also receives encoded messages from upper layers (i.e., what is called the communications stack in Fieldbus) and converts the messages to physical signals that are transmitted over the Fieldbus wire.

Role of Data Link Layer

The Data Link Layer controls the transmission of data and error recovery. The Data Link Layer also controls the rate of information is received from the physical layer. It transmits messages on the fieldbus through the Physical Layer. The Data Link Layer manages access to fieldbus through a deterministic centralized bus scheduler, the Link Active Scheduler (LAS). The LAS controls the exchange of data between devices and schedules transmission of deterministic messages. The Data Link Layer defines these types of devices:

- Basic Fieldbus Device
- Link Master
- Bridges

How FAS and FMS relate to Application Layer

When you first hear the term “Application Layer” it may be easy to think of it as the application itself but recall that this model is concerned with how layers communicate with the User Layer. In reality the Application Layer takes the data sent to it from a user (e.g., an operator setpoint change) or service that sends data, such as reporting an alarm event or broadcasting a PV to a PID control function block. In other words, the application layer prepares it for the other lower layers. In Fieldbus, the FMS and FAS perform the equivalent function to the Application Layer.

2.5 Scheduled Communication Uses Compel Data

Key scheduled communication concept

The key communication concept is that the Link Active Scheduler (LAS) knows which data a Fieldbus device may need and when to make it available for another Fieldbus device. The LAS does this through the use of a compel data message.

Compel data message

The Link Active Scheduler knows which data a device may need to make available to another Fieldbus device. The LAS keeps a list of transmit times for all data buffers (for

example, an AI data buffer) in all Fieldbus devices that need to be transmitted on a scheduled basis. In order for a device to make that value known to another device, it has to send the data buffer on to the Fieldbus network. The device then broadcasts or “publishes” the data for all devices on the fieldbus when it receives a “compel data” message from the LAS.

Publish and subscribe

When a Fieldbus broadcasts a buffered data value, such as an analog input value, it is said to “publish” the data. A Fieldbus device that is configured to receive the data broadcast is called a “subscriber.” Scheduled data transfers using publish/subscribe communications are mainly used in Fieldbus resident control loop strategies.

2.6 Unscheduled Communication Uses Pass Token

Key unscheduled communication concept

Unscheduled communication occurs on Fieldbus to accommodate alarm event, operator display, trend, and other information messaging needs. The Link Active Scheduler grants permission to devices to send unscheduled messages through the use of a pass token message.

Purpose of Pass Token

When the LAS determines that there is time available within an LAS schedule before it has to send out another Compel Data message, the LAS can send out a Pass Token message. If you are familiar with the term “token,” you know that it means that the device, which has the token, has the right to talk on the network. The Pass Token message grants the right to Fieldbus devices to broadcast unscheduled messages such as alarm events, operator display information, trends, and diagnostics.

2.7 Additional LAS Operations

Additional LAS operation concepts

In addition to its scheduling tasks, Link Active Scheduler (LAS) operations include the following:

- adding devices,
- broadcasting time, and
- supporting Link Master redundancy.

How devices are added

When devices respond to a Pass Token message, they are added to a list that the LAS maintain called the Live List. When new devices are installed on the Fieldbus network, LAS uses a Probe Node message to locate them. A Probe Node message is periodically sent to addresses not in the Live List. When a device responds to this message, it is added to the Live List.

How time is broadcast

In order for devices on the fieldbus to perform scheduled communications and execute function blocks at the scheduled time, the devices must have exactly the same data link time. The LAS provides this time when it periodically broadcasts a Time Distribution message on the fieldbus.

How redundancy is supported

Multiple Link Masters provides the LAS with redundancy. If the current LAS fail, another Link Master becomes the LAS and fieldbus communications continue.



ATTENTION

Review your supporting documentation to determine which device performs the role of the Link Master. For example, your system may be one where the host system's interface card performs this role.

3. Fieldbus Wiring and Installation Overview

3.1 Introduction to Wiring a Fieldbus Network

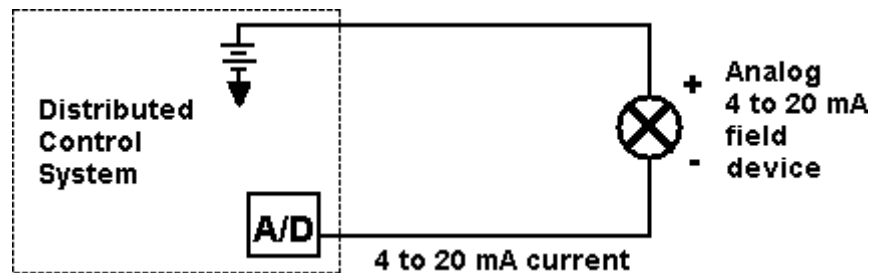
Introductory wiring concepts

Because Fieldbus represents from a wiring and installation viewpoint a two-way, digital, multi drop network with intelligent devices, the basic rules for wiring devices to the network become important. The following introductory concepts are first discussed:

- An installed 4 to 20-ma configuration can be migrated to Fieldbus. A traditional 4 to 20-mA approach and a Fieldbus system are compared to help you migrate to a Fieldbus system.
- A Fieldbus network consists of segments. Both the network and segment have unique concepts that you should become familiar with.

Review a simple network

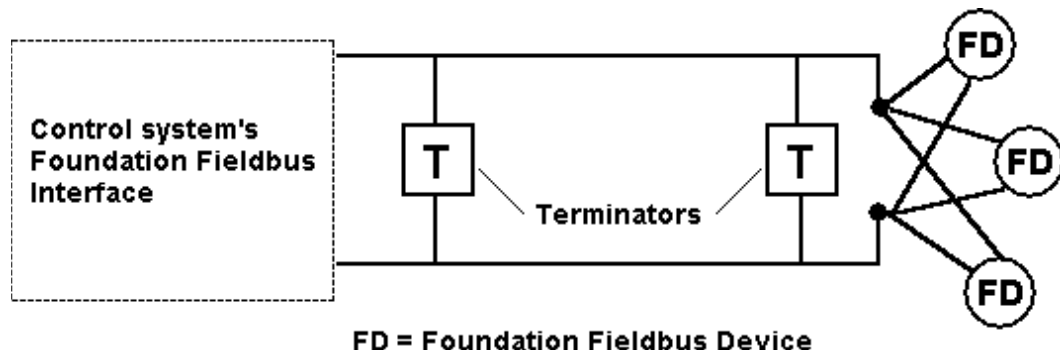
Begin examining a simple Fieldbus network by reviewing a 4 to 20 mA configuration. In a conventional 4 to 20-mA arrangement, two wires usually transmit a signal that represents a process variable to a control system interface.



In Fieldbus, the same principles apply in that two wires transmit a signal to a control system's Fieldbus interface. The Fieldbus field device, which provides many more variables and configuration information than just the process variable, replaces the traditional analog field device. Terminators are required at both ends of the fieldbus segment (as shown in the following figure).

Expand a simple Fieldbus network

You can expand the simple Fieldbus network when you add fieldbus devices in parallel to an existing Fieldbus device.



Fieldbus similarities to 4 to 20 ma configuration

From the previous discussion, observe that Fieldbus has the following similarities to a 4 to 20 ma configuration.

- A twisted-pair cable carries the power and process signal. In both approaches, never ground the wire that carries the signal.
- Existing 4-20 mA wire can be used in a Fieldbus network, although communication wire is preferred (the Fieldbus cable types are described later) in a new Fieldbus network.
- Familiar process devices (transmitters, valves, etc.) are installed in both approaches.
- Both approaches are intended for use by instrument technicians and do not require that the installer become a network communications expert.

Fieldbus differences with a 4 to 20 mA configuration

Fieldbus has the following differences with a 4 to 20 ma configuration.

- Multiple transmitters and valves can be connected to one wire pair.
- Fieldbus devices are connected in parallel, not serial.
- Fieldbus electronics in all devices are different than those found in traditional 4 to 20 mA field devices. For example, the devices can support process control functions within the devices themselves.
- In a fieldbus network, terminators (which consist of a resistive-capacitive circuit) are required to provide good signal quality on each bus segment.
- A fieldbus can have multiple segments, interconnected by repeaters. Repeaters are devices that extend the Fieldbus network's length.

3.2 Fieldbus Components

Key component concepts

A fieldbus network consists of the following components:

- cabling,
- terminators,
- power supplies,
- terminal blocks , and optionally,
- repeaters, bridges, and gateways.

Types of cable

You can use four types of cable for fieldbus wiring. The IEC/ISA Physical layer standard identifies the twisted pair cables as Type A, B, C, or D.

The preferred cable for new installations is Type A. The alternate preferred cable is Type B. Type B cable is likely to be used in both new and retrofit installations when multiple fieldbus segments are run through the same area of the plant. A less preferred cable is Type C and D.

Type	Cable description	Wire size	Maximum length
A	Shielded, twisted pair	(.8 mm ²) #18 AWG	1900 m (6232 ft.)
B	Multi-twisted-pair with shield	(.32 mm ²) #22 AWG	1200 m (3936 ft.)
C	Multi-twisted-pair without shield	(.13 mm ²) #26 AWG	400 m (1312 ft.)
D	Multi-core, without twisted pairs and having an overall shield	(1.25mm ²) #26 AWG	400 m (1312 ft.)



ATTENTION

The preferred cable for new installations is Type A.

Terminators

In general most of today's networks require terminations to provide good signal quality. The usage of terminations is a requirement for a Fieldbus network to work properly. If a Fieldbus network does not have terminations, then when a signal is sent on to the wire medium and an open wire (or short) is encountered, it is very likely that the signal will bounce back as an echo. The echoed signal is called a reflection; the reflection distorts the original signal.

For a Fieldbus segment, TWO (and ONLY TWO) terminators are needed. The terminator consists of a specified resistive and capacitive configuration, but you do not need to be concerned about assembling your own terminator.



ATTENTION

Terminations come in various configurations - - they could be incorporated within the terminal blocks, IS barriers, power supply, or host system interface card.

Power supplies

The Fieldbus Foundation specifies several power supply types (for intrinsically safe and non-intrinsically safe applications). It's important that you use a power supply impedance matching network in fieldbus networks. The impedance matching network is a resistive/inductive network that is either external or built into the fieldbus power supply.

Repeaters, bridges, gateways

Your fieldbus segment can accommodate special communication devices such as repeaters, bridges, and gateways. A repeater, bridge, or gateway appears as another device on your fieldbus segment.

A brief description of how the devices work follows.

- Repeaters extend a network.
- Bridges interface to different cable types or speeds.
- Gateways interface to different communication protocols.

Repeaters are considered active devices (either bus powered or non-bus powered) that extend your fieldbus network. Up to four repeaters can be used between any two devices on a fieldbus network.

Bridges also are active devices (either bus powered or non-bus powered) that are used to connect fieldbus segments of different speeds or cable medium (e.g., fiber optic, twisted pair) effectively forming a larger fieldbus network.

Terminal blocks

Terminal blocks currently used with 4 to 20 ma signals can be used with your fieldbus wiring. These terminal blocks support multiple bus connections. Additionally there are terminal blocks especially designed for fieldbus applications that make wiring more convenient. The fieldbus blocks have internal connections that help avoid polarity errors. The termination methods include screw terminals, spring clamps, and pluggable connectors.

What does H1 represent?

The Foundation Fieldbus originally had intentions of specifying both a low speed and high speed network implementation. The low speed specification is called H1. H1 defines a transmission speed of 31.25 thousand bits per second (kbps). The H1 specification is appropriate for most process control applications such as temperature, flow and level. It is considered too slow for many discrete control applications that require greater response speed. (The initial high-speed specification, H2, is now replaced with an effort to define a High Speed Ethernet, HSE, specification.)

Network basics

The following **network** basics are true of fieldbus systems.

- A fieldbus network (also called a link or bus) consists of 1 or more **segments**.
- All physical and electrical (the total resistance, their power requirement) properties apply to **segments**.
- All logical properties (Link Active Scheduler) apply to **networks**. For example, the Host device containing a Link Master schedules the network but only powers one segment.
- A network must have at least one Link Master. If there is more than one Link Master, at any point in time only one Link Master becomes the Link Active Scheduler (LAS).
- An H1 network has a bandwidth of about 50-scheduled messages/second.
- Bridges can interconnect networks. Because a bridge contains a buffer, messages between devices on different networks will have a fixed delay.

Segment basics

The following **segment** basics are true of fieldbus systems.

- A segment is one wire pair with exactly 2 terminators. Review your system documentation to determine whether terminations are supplied within the terminal block(s) or host system interface.
- A segment may have exactly one power supply load. The power source is either from the host control system or external power supply.
- Each device on a segment must receive at least 9v. Review each device's power requirements as they may vary.
- A segment may have one Intrinsically Safe I.S. barrier; it must be within 100m of a terminator. Barriers limit current, and thus limit the number of bus-powered devices.
- A repeater counts as one fieldbus device.

3.3 Wiring Topologies

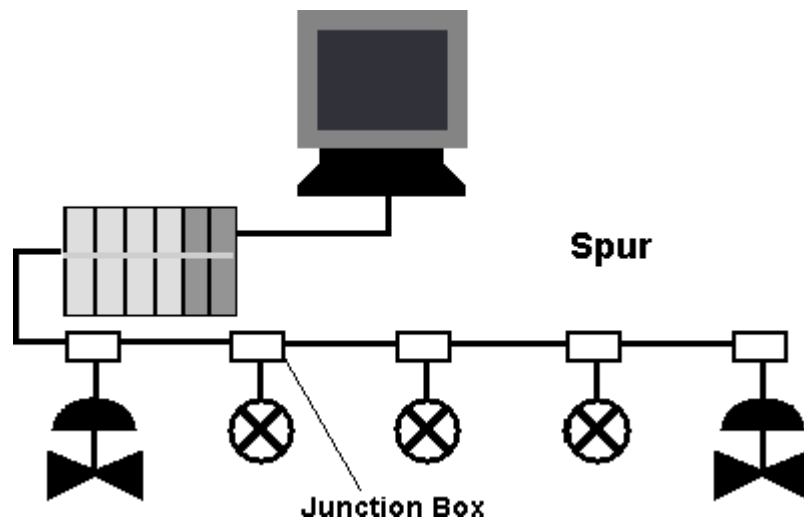
Key topology concepts

Fieldbus networks can appear in several different wiring topologies. Topologies refer to the architecture or physical configuration of the network devices, cabling, and network components such as terminations. Some common topologies include

- bus with spurs,
- daisy chain, and
- tree.

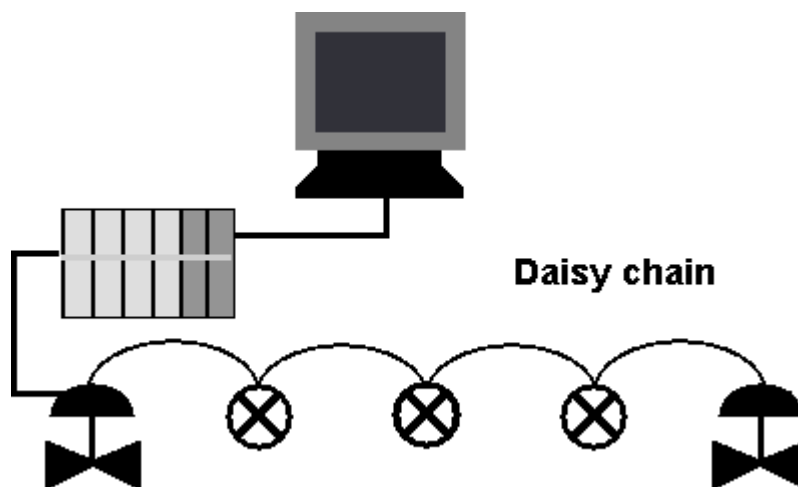
Bus with spurs topology

In the bus with spurs topology, fieldbus devices are connected to the bus segment (also known as the home run cable) using a length of cable called a spur.



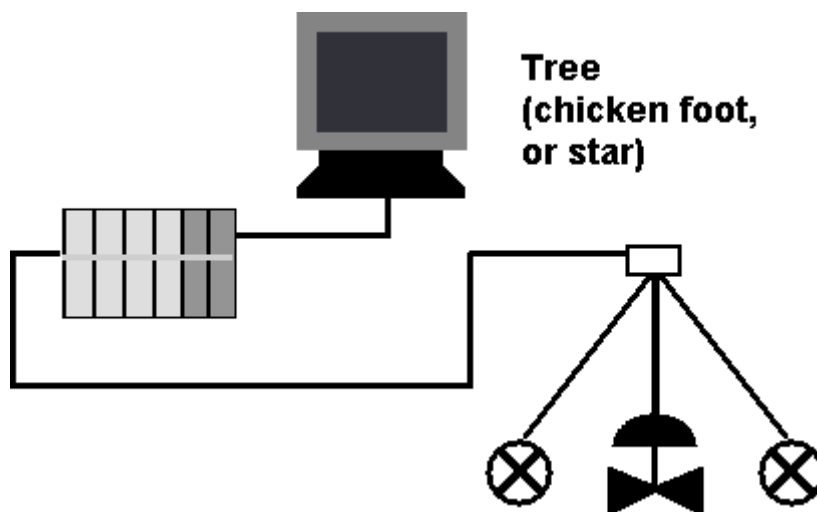
Daisy chain topology

In the daisy chain topology, fieldbus devices are interconnected at the terminals of each device, the fieldbus cable is routed from device to device.



Tree topology

In the tree (also known as a “chicken foot” or star) topology, fieldbus devices are connected at a common termination assembly. The assembly can be a junction box, terminal, marshalling panel, or I/O interface. The assembly is often located at the end of a bus segment (also known as a home run cable).



3.4 Summary of Fieldbus Grounding Rules

Key grounding concepts

The following are key grounding concepts.

- Fieldbus signals are not connected to ground in any device.
- Fieldbus cable shields are
 - most effective when grounded at only one place, and
 - usually grounded in the control room.
- Surge Protection
 - may be included in Fieldbus wiring blocks, and
 - surge suppressors are separately grounded at each wiring block.

Signals NEVER grounded

One of the most important grounding rules to remember is that you should NEVER connect either conductor of the twisted pair wire to ground at any point in the network. The reason for this rule is that the fieldbus signal is sent throughout the network and grounding either conductor can cause ALL devices on the network to lose communication.



WARNING

NEVER connect either conductor of the twisted pair wire to ground at any point in the network. Grounding a signal wire can cause ALL devices on the network to lose communications.

Grounding cable shields

Cable shielding is intended to keep noise from interfering with the fieldbus signal. The cable shield is most effective when it is grounded at only one point along the cable. If the cable is grounded in more than one point, it is possible to introduce ground currents that cause noise on the twisted pair conductors. Grounding the cable shields often occurs in a central location such as a control room. At the segment itself, the grounding typically occurs at the power conditioner or at an intrinsic safety barrier.

Surge protection approach

Because outdoor fieldbus wiring risks being exposed to lightning strikes or large currents, surge suppressors are used. At high voltages, the surge suppressor becomes part

of a low resistance path to ground. Surge suppressors are built into some terminations and terminal blocks. When used, the surge suppressors are separately grounded at each wiring block.

3.5 How Many Devices are Allowed on H1?

Key device limit concepts

Although Fieldbus can support a large number of devices on its network, it all depends on the power supply capacity, length and resistance of cable, bandwidth and whether the network is intrinsically safe. When considering practical limits, the number of fieldbus devices is limited to a smaller number than the absolute technical limits.

Non-intrinsically safe device limits

The number of devices on a non -intrinsically safe network depends on the following factors.

- **Bandwidth** - A fieldbus can carry approximately 50 messages/sec. There could be, for example, 5 devices at 10 messages/sec each or 200 devices at 1 message at 4 seconds each.
- **Segments** - A fieldbus can have 1 or more segments of up to 1900 meters each, with up to 32 devices per segment. If repeaters are installed, then you can a maximum of 240 fieldbus devices.
- **Power** - The power supply on each segment must have sufficient capacity. Devices draw different amounts of power; some draw none because the devices are separately powered.
- **Wire Gauge** - Existing 4 to 20 mA wire with high resistance may reduce the segment length.

Intrinsically safe device limits

The number of devices on an intrinsically safe network depends on the following factors.

- **Bandwidth** - A fieldbus can carry approximately 50 messages/sec (same device limits as a non-intrinsically safe network).
- **Segments** - A fieldbus can have 1 or more segments of up to 1900 meters each, with up to 32 devices per segment. If repeaters are installed, then you can a maximum of 240 fieldbus devices (same device limits as a non-intrinsically safe network).
- **Power** - The Intrinsic Safety (I.S.) barrier must be within 100 meters of a terminator. The amount of power available past the I.S. barrier is approximately 800 mw.

Because devices draw different amounts of power, you have to account for their total power requirements. For example, you could have eight 100mw devices or four 200mw devices. As a rule of thumb consider 2-6 devices past I.S. barrier.

- Wire Gauge - Existing 4 to 20 mA wire with high resistance may reduce the segment length (same device limits as a non-intrinsically safe network).

Technical device limits

The following are technical limits regarding the number of devices per segment.

- A segment may have 32 device loads.
 - For example, you could have the following segments. Segment 1 has 8 loads (1 DCS interface and 7 repeater loads). Segments 2 through 7 have 2 loads each, each repeater and device counts as a load
- There can be no more than four repeaters in series between any two devices.
 - For example, you could have one repeater between the host control system (e.g., a DCS) and any device, or two repeaters between any two devices.
- A fieldbus may have up to 240 Devices.
 - Many more than 240 devices could be properly connected, but only 240 can be addressed.

Practical device limits

From a practical standpoint, the number of devices on a segment is much less than the technical given a particular plant's fault tolerance for control approach and wiring topologies.

- Fault tolerance for control
 - Some plant applications cannot tolerate the loss of more than one loop due to a single failure.
 - For that reason, a fieldbus segment would have devices for just one loop: 1 valve and 1-2 transmitters. Some monitoring devices could be added to that segment.
- Wiring topology
 - Plant wiring typically runs from a device to a junction box to the control room.
 - When devices are close together, wiring them to a single bus is convenient.

- When devices are far apart, a single bus may require additional wire runs.

In summary, a plant's process control fault tolerance, topology, power, and barriers/isolators may limit the number of devices at 3 to 8 devices per segment. If the devices are monitoring only devices, additional devices can be considered.



ATTENTION

Be sure to review the Honeywell documentation for specific device limits for the control system you are connecting to. For example, you may encounter specific guidelines limiting the number of devices to ten per segment with two of the devices being final output devices.

3.6 Pre-existing Wire Installations

Key pre-wired cable concepts

Wire that is already installed can be reused to build a new segment provided the total length of the wire in the segment is within the specified limits for Cable Types A, B, C, and D. Wire that does not fall within the specified limits should not be reused.

Calculation

The calculation to plan the segment as if the existing wire is new wire is the following:

$$LX/MAXX + LY/MAXY < 1$$

Where LX = length of cable X

LY = length of cable Y

MAXX = maximum length for X alone

MAXY = maximum length for Y alone

Example calculation

An example calculation follows. Assume in our example that LX represents 1100 meters of type A cable mixed with 100 meters of type D cable. Your calculation would be worked as follows:

LX = 1100 m cable type A

LY = 100 m cable type D

MAXX = 1900 maximum length for type A alone

MAXY = 200 maximum length for Y alone

$$1100/1900 + 100/200 < 1$$

1.079 is not < 1. Because the result is >1, a different cabling solution must be found.

Expanded formula

The cable calculation formula can be expanded to include the four cable types.

$$LX/MAXW + LY/MAXX + LX/MAXY + LY/MAXZ < 1$$

Where LW = length of cable W

LX = length of cable X

LY = length of cable Y

LZ = length of cable Z

MAXW = maximum length for W alone

MAXX = maximum length for X alone

MAXY = maximum length for Y alone

MAXZ = maximum length for Z alone

3.7 Calculating DC Power for Two Wire Devices

Key DC power concepts

The number of devices on a Fieldbus segment is a factor of the power supply voltage, the resistance of the cable, and the amount of current drawn by each device.

Calculation data

Use Ohm's Law to determine the amount of current that is available for the field devices. You need to know your power supply voltage, the resistance of the total length of the cable and the amount of current your devices draw.

Example calculation

The following example shows how to use Ohm's Law to calculate the number of devices allowed on a segment. In our example, assume a power supply and power conditioner output of 19 volts (nominal). Next the cable resistance is based upon cable type and length. Assume that we are using a 1km length of 18-gauge wire. It has a resistance of 22 ohms per kilometer. Because there are two conductors, the 1km length has a resistance of 44 ohms. Assume also that each device requires 20 ma.

Use this information to calculate total current

Voltage/resistance = Current

10 Volts/44 ohms = 227 mA

Because the devices require 20 mA, the total number of devices is 11, based upon 227 mA available/ 20 mA per device = 11 devices.



ATTENTION

Remember that you also have to consider any additional limits imposed by barriers, host system interfaces, or plant's fault tolerance strategies.

Test for right voltage and connectivity

You can use an ordinary digital voltmeter to test for the right voltage at the wiring of each device. Additionally there are testers available from third party sources that can verify the signals on the network as well as network testers.



TIP

For best maintenance preparedness, your maintenance department should have available the following

- A digital voltmeter
 - A Fieldbus wire tester
 - A Fieldbus network tester
-

Power requirements will vary

Our earlier example illustrating power calculations with Ohm's Law was relatively simple. You should note that the power requirements for fieldbus devices vary by device type and manufacturer. Additionally, power calculations can become quite involved when a number of devices are located on the cable in different locations and topologies. Honeywell documents provide examples of these calculations; software programs are also available to automatically perform the device loading calculations.

3.8 Polarity Considerations

Key polarity concept

It is best to observe polarity when wiring Fieldbus devices, even though some Fieldbus devices are non-polarized.

Treat network as polarized

The signal that a Fieldbus device looks for is one of an alternating voltage. Because positive and negative voltage signal swings have different meanings, the fieldbus signal is polarized. Fieldbus devices must be connected so that they can see the signal in the correct polarity. If the Fieldbus device is connected with the wrong polarity, the device may short out the network or just not be able to communicate.

Observing polarity simply means that all of the (+) terminals must be connected to each other. Likewise all of the (-) terminals must be connected to each other. Although some Fieldbus devices are non-polarized (i.e., they can be connected in either polarity), it is best to observe polarity. Color coding conventions help in wiring the devices properly. Although the Fieldbus standard does not specify the wire colors for the (+) and (-) terminals, one approach for new installations with twisted pair cabling is to use orange wire for the (+) terminal connections and the blue wire for the (-) connections.



TIP

Consider all devices as polarized, then you do not have to worry about which devices are polarized and which devices are not polarized.

Bus powered device polarity

Field device receiving power from the bus segment may also appear in polarized configurations. Because these devices have the signal and power polarity designed for the same terminal, correct polarity is made when you connect the + DC to the + terminal.

